Department of Economics

# MSc Extended Essay Cover Sheet

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*Double spaced, typed. Maximum 6,000 words* (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered. Binding is not necessary, however it is crucial that all pages are secure, i.e. stapled.*

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# How does Job Flexibility impact Child Penalties?

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EC423 - Labour Economics

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**Abstract**

This paper explores how job flexibility impacts child penalties in order to explain differences observed before and after the COVID pandemic in the United Kingdom. Through event studies that leverage the Dingel et al work from home flexibility measure on panel data in the United kingdom, I find that women in flexible occupations 1 year before childbirth show a 30% drop in labour income whilst women in non-flexible occupations show a 40% drop. Controlling for income shows that flexibility protects women in lower income the most. Analyzing hours worked, participation, and wage rates as channels through which labor income is affected reveals that wage rate is the primary channel followed by participation.

**Word Count :** 6,330 words

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# 1 Introduction

Despite women converging towards men in terms of labour force participation, hours worked, wage rates and surpassing men in terms of education attainments, inequality in earnings persists (Goldin 2014). This lingering inequality is increasingly explained by lifecycle effects, particularly motherhood (Kleven, Landais, and Søgaard 2019a). Mothers see their income, labour force participation, hours worked and wage rates fall sharply after the birth of their first child and fail to recover. Fathers, however, do not see a similar drop in any of these measures. The relative decline compared to that of the father following birth is what Kleven et al coined the child penalty. The incredible persistence of these penalties across time and across countries can be partly explained by social norms surrounding household organization, that is who takes on the burden of childcare, which have yet to converge... (Ferrant et al. 2014) (Kleven, Landais, and Søgaard 2019a)

While societal norms drag behind, the COVID pandemic forced a surge in work from home arrangements and as zoom meetings became the norm, household organization may have changed as well (Aksoy et al. 2023)(Davis 2024). This shift raises a twofold question: How does job flexibility impact child penalties and how have these child penalties evolved following the pandemic? My question situates itself within three areas of the literature. The first area focuses on research around the child penalty, the second on job flexibility and its impact on mothers' labor supply, and the third on the impact of COVID on women's labor supply.

There has been a great deal of papers replicating the child penalty method set out by Kleven et al to other countries and demographic groups (De Quinto et al. 2020 ) (Andresen and Nix 2022) (Kleven, Landais, and Søgaard 2021 )(Kleven, Landais, Posch, et al. 2019) (Liu 2021). A particularly relevant study to my question finds child penalties in the UK using the same dataset I use (Andrew et al. 2021). Their paper also finds that part-time work penalties are higher for more highly educated women. This suggests

that there might be occupation specific characteristics that hinder some mothers more than others, and motivates our approach to look at flexibility as a characteristic.

The theory of occupational pay differences explains how occupations might differ in their returns to hours worked. To put it simply, some occupations might reward workers in a quadratic fashion while some increase linearly in hours worked. In her research, Claudia Goldin finds that the Pharmacy occupation is one that increases linearly in hours worked (Goldin and Katz 2016) whereas lawyers is one that increases non-linearly (Goldin 2014). Consequently, income penalties for reduced hours following childbirth are lower for pharmacists than for lawyers. In terms of the theory, since lawyers are rewarded more for extra hours, reducing hours following childbirth penalizes them more than pharmacists. Supporting this theory, a paper specifically looking at child penalties by sector in Belgium finds that sectors with atypical work schedules or irregular hours have greater penalties (Sébastien Fontenay and Tojerow 2023).

In order to look at occupational differences researchers have created different job flexibility measures. Claudia Goldin defines occupational flexibility using O*NET characteristics as one that does not have strict deadlines, requires less face to face meetings, does not require social relationships to be maintained, and is less substitutable. With these metrics, the technology sector is the most flexible and also shows smaller part-time penalty for new mothers (Goldin 2014). Research applying this measure to evaluate the graduate gender wage gap in the UK finds that women tend to move to more flexible occupations over their lifecycle, whereas men do not, which helps explain the gender gap (Benny et al. 2021). In order to evaluate changes related to covid, my essay measures job flexibility as the ability to work from home. The main difference in this measure is that it does not consider substitutability and has a stronger emphasis on locational flexibility, that is how much work needs to be done at the employers premises. As such, more physically demanding occupations in construction or nursing are coded as less flexible (Dingel and Neiman 2020). Research using this index focused mostly on explaining how the pandemic exacerbated inequalities as people with lower incomes

tended to be in less flexible occupations, a finding my research also supports. Additionally, research also finds that higher income countries have a greater proportion of flexible jobs (Gottlieb et al. 2021).

The impact of COVID on women's labor force has also been well documented particularly in the US. On the shorter term, research shows that women's labor force participation in terms of hours worked decreased relative to men's (Collins et al. 2021). When effects are disaggregated, mothers with school age children were found to be more impacted by the pandemic as they took on increased childcare responsibilities due to school closures, while fathers were less affected. Importantly, they find that women with younger children were not as impacted (Fairlie et al. 2021). Since my event study window is two years pre and post first childbirth, mothers in my sample are not affected by school closures. Although they might be affected by other childcare services like nursery closures. In their decomposition exercises, Couch et al also find that being able to telework might have mitigated child penalties (Fairlie et al. 2021).

By leveraging the flexibility measure outlined by Dingel to analyze child penalties, my paper explores the intersection of job flexibility, child penalties and the effect of COVID on women's labour supply in a novel way. In this paper, section 2 presents the data and sample selection, section 3 describes the model and identification strategy, section 4 computes child penalties before and after COVID, section 5 introduces the flexibility index, section 6 computes child penalties by income and flexibility groups, section 7 discusses limitations and possible extensions and section 8 concludes.

## 2 Data

The analysis is based on Understanding Society, the UK Household Longitudinal Study (UKHLS) (University of Essex, Institute for Social and Economic Research 2023). It is a longitudinal survey of approximately 40,000 households (in wave 1) in the United Kingdom. The data spans over 13 waves, from 2009 to 2023, waves are not exactly one year. The survey contains high-quality longitudinal data on subjects such as health, work, education, income, family, and social life. I extract metrics such as hours worked, income, wage, occupation codes all linked to individuals via personal identification numbers.

My event studies are centered around the birth of the first child with parents who are observed 2 years prior and 2 years after childbirth. Thus, I can only observe parents who had their child between 2011 and 2021. I did not restrict my sample in terms of marital status, which allows for larger samples but can create positive bias which I discuss later on. I chose my COVID sample to be parents who had their child between 2019 and 2023. The final sample is described in the table below. I realize that the sample is small compared to studies that use administrative data, especially for the COVID sample. The panel is not balanced in terms of number of child births observed per year. The maximum number of child births in a year was 226 births, the smallest in 2021 was 2 births. The two time periods are also not identical in terms of individuals since they include a different number of waves. Table 1 summarizes my final samples.

| | Mothers | Fathers | Total |
|---|---|---|---|
| Total sample of parents (-2 to 2) | 909 | 712 | 1621 |
| Sample of parents (2009-2018) | 792 | 623 | 1415 |
| Sample of parents (2019-2023) | 117 | 89 | 206 |

Table 1: Sample descriptives

I estimate child penalties using income, labour force participation, hours worked and wage rates. Participants responded to the survey yearly and were asked about their weekly hours worked and monthly income from different components (savings, labour, investments, benefits). For my analysis I used the labour component of income (commonly referred to as earnings). I subsequently refer to this variable as income. Wage rates are calculated as monthly income divided by weekly hours worked (multiplied by 4.14 to make them monthly) for those who are participating in the labour force. There are several limitations to my data other than its size. First, because the survey is only done yearly, the hours measure is not very precise and does not allow us to see changes such as work interruptions or switches to part-time work within a given year. Also, because it is a survey there is considerable concerns of measurement error. The wage rate variable is especially impacted since weekly hours are transformed into monthly hours. To that end, I decided to calculate wage rate penalties for those who had worked above 10h weekly. This reduced outliers that underestimated child penalties.

# 3 Identification and Event Study approach

To identify the effect of childbirth on the outcomes I make several assumptions. The first essential assumption in the event study is that the event, in the case childbirth, is exogenous to the outcome variable. Childbirth is not determined by hours worked, wage rates, or income. This allows for fertility to be endogenous, we are only looking at the birth of the first child not of future children. This assumption is supported graphically in my event studies by noticing that prior to birth, the outcomes are not responding (see figure 1). A limitation to the identification strategy is that it cannot capture pre-child effects such as career or educational choices of mothers anticipating a child. Since these pre-child effects are differenced out in the event study as such, I can only capture post child effects. Another assumption that is made, which is related to the first, is that the average non-child earnings path would have been identical for women who gave birth and men who did not, controlling for covariates. A point of departure from the

Kleven paper is that their event study captures very long run effects (up to 20 years post child). To that end, their specification included the issue of total lifetime fertility not just the birth of the first child. Since my event window is smaller, the event study is less prone to this bias. The initial model specification, which I explicit below, is the same as the Kleven et al paper (Kleven, Landais, and Søgaard 2019a) (Kleven, Landais, and Søgaard 2019b).

$$Y_{ist}^{g} = \sum_{j \neq -1} \alpha_j^g \cdot \boldsymbol{I}[j = t] + \sum_k \beta_k^g \cdot \boldsymbol{I}[k = age_{is}] + \sum_y \gamma_y^g \cdot \boldsymbol{I}[y = s] + \nu_{ist}^g$$

In this model, $Y_{ist}^{g}$ denotes the outcome: income, hours worked, participation and wage rates. It is indexed by person i, event time t, year s and gender g. The first term on the right hand side, is the event-time dummies, the second is the age dummies, the third is year dummies. Year dummies capture changes in outcome due to transitory effects such as inflation and other business-cycle effects. Age dummies are included to capture life-cycle effects. This would capture effects that are generally seen in the population that would otherwise show up in the event estimate. The event time -1 is excluded so that the measured results are all relative the pre-birth year. The regression outlined above is run separately on mothers and fathers using robust standard errors. Since the interpretation compares fathers to mothers implicitly I discuss heterosexual couples. However, research looking at gay and lesbian couples found that lesbian mothers still experience a child penalty while gay fathers do not (Andresen and Nix 2022). Outcome variables are not transformed using logs to be able to include participation. Then, using the same notation, the predicted outcome without the event dummies, what would have happened without a child, is specified below.

$$\hat{Y}_{ist}^{g} \equiv \sum_k \hat{\beta}_k^g \cdot \boldsymbol{I}[k = age_{is}] + \sum_y \hat{\gamma}_y^g \cdot \boldsymbol{I}[y = s]$$

The within gender penalty is defined as the year-t effect of having children $\hat{\alpha}_j^g$, estimated in our first regression, as percentage of the estimated counterfactual.

$$P_t^g \equiv \frac{\hat{\alpha}_t^g}{E[\hat{Y}_{ist}^g|t]}$$

Finally, the child penalty at time t, is defined to be the percentage difference of the effect of having a child on a man versus a woman, using the following specification:

$$P_t \equiv \frac{\hat{\alpha}_t^m - \hat{\alpha}_t^w}{E[\hat{Y}_{ist}^w|t]}$$

In order to evaluate the effect of job flexibility on child penalties, I estimate the penalties as above but run separate regressions by flexibility groups that I specify in section 5. In order to differentiate between my COVID and non-COVID samples, I also run separate regressions on both samples. When separating my samples by flexibility, the identification story changes as partners are not necessarily in the same flexibility group. I discuss later limitations to my approach.

# 4 Child penalties before and after COVID

I begin the analysis by calculating child penalties before and after COVID. In the pre-COVID sample, mothers face significant gender penalties in income, hours worked, and labor force participation, while fathers experience minimal penalties across all metrics (see figure 1). Controlling for age and year effects reveals similar pre-trends for both genders, reinforcing our earlier exogeneity assumption.

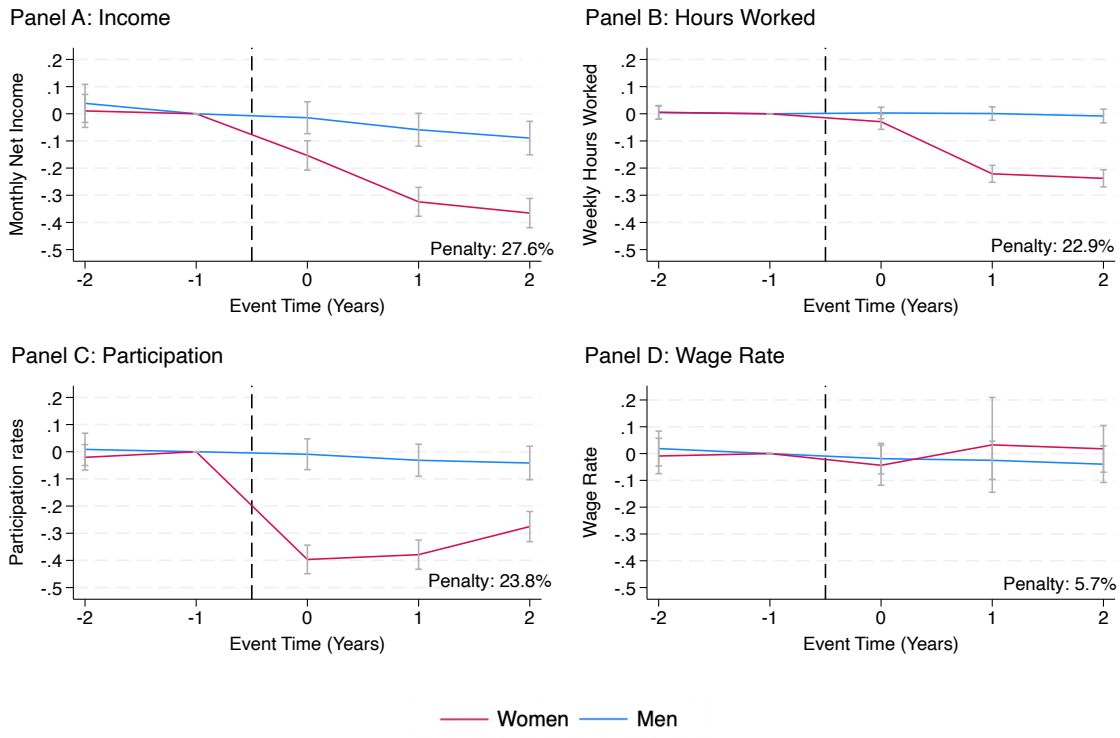


Figure 1: Child penalties for 2008-2018 sample

Note: All penalties are calculated relative to t=-1. The figures plot the $P_t^g$ estimated for each gender at every event time using the regression described above. The bars represent 95% confidence intervals using robust standard errors. On the bottom right of each panel are the child penalties $P_t$ calculated at t=2. Panel A observations include 3115 observations for men and 3959 for women. Panel B has 2472 observations for men, 2584 for women. Panel C has 3115 observations for men and 3959 for women. Lastly, Panel D has 2472 observations for men, 2584 for women.

Panel A, which illustrates labour income penalties, finds no gradual convergence of women towards men in our time frame. Two years post childbirth, women are earning 27.6% less relative to men. I find that the decline in income is progressive as opposed to immediate following childbirth. This is a point of departure from results seen in Denmark in the Kleven et al paper who find that the labour income penalty peaks 1 year post childbirth and starts converging until reaching a plateau at around 2 years post childbirth (Kleven, Landais, and Søgaard 2019a). However, my estimates are in line with Andrew et al, who find a similar decline in the UK until 2 years post birth in levels rather than percentages, which suggests country specific trends (Andrew et al. 2021). The drop in labour income can come from three channels. Mothers could work less in terms of hours worked (the intensive margin), stop working altogether

(the extensive margin), or their wage per hour might have dropped. I examine these channels next.

Panel B, which illustrates penalties in hours worked, shows that mothers decrease their hours worked after childbirth, reaching a 22.9% penalty relative to men two years after childbirth. Interestingly, mothers are not significantly reducing their hours worked before and at childbirth, another point of departure from mothers in Denmark who decrease their hours worked immediately at childbirth. The drop in income at childbirth is rather explained by Panel C, which looks at participation. On this margin, mothers' participation drops by 40% at the event time. The difference between the UK and Denmark in the timing between hours worked and participation could suggest differences in maternity leave policies which I do not explore here. There is also convergence in this panel which suggests women are gradually reentering the labour force, decreasing the penalty to 23.8% two years after childbirth. A comprehensive study of child penalties in employment level across countries, the child penalty ATLAS, finds a 34% penalty for the UK. The ATLAS estimates have been re-scaled for cross-country comparisons and uses a more comprehensive labour force survey which could explain differences (Kleven, Landais, and Leite-Mariante 2023).

The last panel is the most perplexing as it suggests wage rates of women are not impacted by children, reaching a penalty of 5.7% two years after childbirth. While my estimates do not fit with the Kleven et al estimates they are similar to the study on UK data who find no significant differences of wages rates in levels (Andrew et al. 2021). This suggests that the income penalty in the UK is primarily driven by a decrease in labour force participation on the intensive and extensive margin which is particularly motivating since job flexibility could potentially act on these two margins.

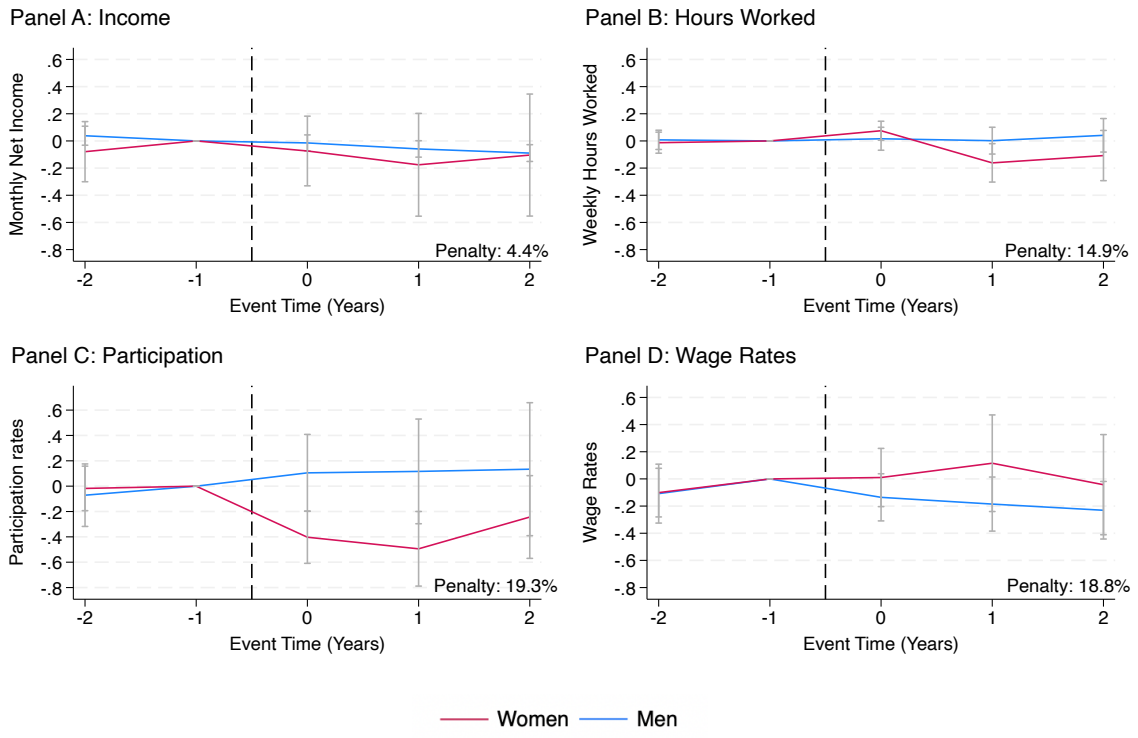


Figure 2: Child penalties for 2019-2023 sample

Note: All penalties are calculated relative to t=-1. Panel A observations include 445 observations for men and 585 for women. Panel B has 343 observations for men, 380 for women. Panel C has 445 observations for men and 585 for women. Lastly, Panel D has 343 observations for men, 379 for women. One outlier was excluded from panel D.

Figure 2 replicates the analysis for the COVID sample of parents. Overall, I do not find significant results at the 5% level due to the small sample size. However, in terms of trends, I find a significant departure from the pre-COVID event studies in terms of wage rates and income. Child penalties in income are 4.40% after two years while child penalties in terms of wage rates show fathers incurring a 18.8% penalty relative to mothers. Interestingly, the significant departure from pre-COVID trends is caused by the decrease of fathers' wage rates rather than an increase in mothers' wage rates. While I do not look at changes in occupation the gradual decline in wage rates could suggest that during COVID, men changed to lower paying occupations following childbirth. Then, the lack of income penalty is partly explained by the compensation in wage rates as participation rates show a similar trend to pre-COVID estimates reaching a penalty of 19.3% two years after childbirth. On the intensive margin, I find similar

trends to the pre-COVID estimates although child penalties after two years are 8% smaller than pre-COVID estimates. While the interpretation is severely limited, the lower penalty in hours could suggest a shift for those on the intensive margin, which further motivates our approach of looking at flexibility.

# 5 Child penalties and flexibility

In this section, I first describe the flexibility index I use, how it is calculated and transposed to the UKHLS dataset. I then summarize some descriptive statistics of occupation rankings based on flexibility and calculate child penalties using the new index. Due to limitations in my data, I focus my analysis on the pre-COVID sample assuming findings in terms of flexibility should transpose to post-COVID era. However, since work arrangements have considerably shifted during COVID, it might be that underlying mechanisms such as occupational pay differences have as well which limits this transference.

## 5.1 Building the Flexibility Measure

To build the flexibility index, I use a similar approach to the paper by Dingel et al who created a work from home measure using O*NET data (Dingel and Neiman 2020)(O*NET 2015). The O*NET database gathers hundreds of occupation characteristics for almost 1,000 occupations. The data is collected through surveys asking about various job descriptors. An example of question: "How often does your current job require you to work outdoors, exposed to all weather conditions?" The respondent would then select a number from 1 (never) to 5 (everyday). I used the same characteristics as the Dingel measure extracting a total of 43 characteristics. These include "Exposed to diseases or infection weekly" or "Handling and Moving Objects". The 43 characteristics are found in two broad surveys. The work context survey which asks questions about work setting, its possible hazards, the pace of work, and dealings with other people,

and the work activities survey which asked about actions that are performed for the occupation.

Since I focus my analysis on the pre-COVID sample, I use the 2015 O*NET database rather than the most recent one. After extracting the characteristics, I followed the method outlined in the Dingel paper which gives a binary index for each characteristic by occupation. I keep my final index non-binary by taking the mean over all the characteristics for each occupation which enables me to do different occupation groups later on. Then, an occupation with a flexibility ranking of 1 (least flexible) would have scored high for all 43 characteristics. The flexibility scores I obtain range from .38 to .03. Finally, I transpose the occupations and their flexibility index to occupations in the UKHLS dataset. For this, I used crosswalks to move from US SOC 2010 classification of occupations, used in the 2015 O*NET database, to ISCO88 which is an international classification used in my dataset (Hardy 2016). While my index estimates are not the same in terms of numbers as the Dingel index, the occupations that are coded as flexible or not are very similar. A limitation of this index is that occupations tend to be coded as more flexible than they would be in actuality. For example, secondary school teachers is one that is coded as being flexible. Another limitation is that I do not weigh how much each characteristic matters for a specific occupation. So while the rankings help determine the importance of an individual characteristic, when taken as a whole they all count the same towards the flexibility index.

Using the new flexibilty measure on the UKHLS dataset, I find that women tend to be in more flexible occupations compared to men. Their average index being 0.109 and 0.126 respectively. However, the top occupation for mothers, which contains 10.23% of the mothers in the non-covid sample: "Personal care and related workers" is fairly non-flexible 0.15 in terms of our index. While the top occupations for mothers are relatively similar across samples, their proportions differ which also limits the extent to which the analysis can be extended between the two periods. Although, this is mostly due to the fact that there are less people and types of occupations in the Covid sample. Men's

top occupations include "Computing professionals", "Other departmental managers" and "Finance and sales associate professionals", with associated index of 0.07, 0.06 and 0.07 respectively. We see a similar decline in the variety of occupation pre and post covid and a drop in the sample. More detail about these descriptives can be found in the appendix.

## 5.2 Calculating Child penalties by flexibility

Since the amount of individuals in each occupation is limited, I create 4 groups based on quartiles of the flexibility index which allows me to calculate child penalties on larger sub-samples. The appendix contains information on top occupations in this new occupation grouping. I calculate child penalties for the precovid sample in the same way as before but separate the sample by the occupation grouping of the parent a year prior to birth. The reasoning being that occupational flexibility before having children might mitigate some of the child penalties. Here, I am comparing parents that are in the same flexibility occupation group, meaning that this does not necessarily show me dynamics within couples if they are in different flexibility groups. Child penalties then represent the percentage by which mothers are falling behind fathers in the same occupational flexibility group due to children. To that end, I place more emphasis on gender penalties, which show effects for one gender rather than child penalties which compare both genders. Figure 3 shows these penalties in terms of income for each flexibility group.

Overall, the flexible occupations are associated with lower penalties for mothers, around 30% for mothers at t=2 compared to 40% for less flexible occupations. However, we no longer see smooth penalties for fathers. Men seem to incur a penalty of around 20%, except for the most flexible occupations. The decline in income for men makes it so that the most flexible occupation actually have the highest child penalty of 31.9% two years after childbirth. That is, mothers are falling behind more relative to men

in those same flexible occupations. So while flexibility seems to damper the impact of children on women, it also seems to have that effect on men thereby decreasing the convergence between genders.

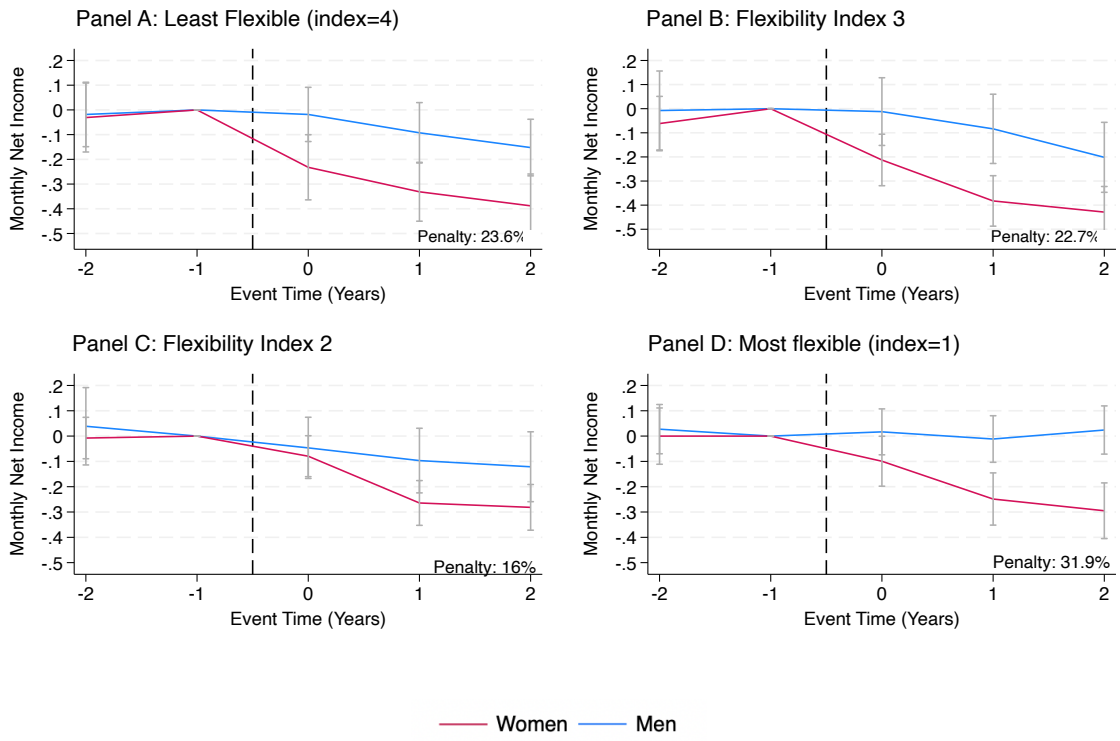


Figure 3: Child Penalties in labour income for the Pre-COVID sample by flexibility index quartile

Note: Panel A observations include 650 observations for men and 865 for women. Panel B has 565 observations for men, 1044 for women. Panel C has 570 observations for men and 890 for women. Lastly, Panel D has 795 observations for men, 815 for women.

A channel that I cannot observe are dynamics within couples. Such a dynamic, which would explain the penalty for men, is that men in inflexible occupations are incurring a penalty by either decreasing their participation or due to lower wage rates. A major issue in detangling these effects comes from the fact that income and job flexibility might be correlated and might be correlated differently based on gender. In fact, the literature suggests that there is a strong relationship between jobs that are able to be done at home and income (Dingel and Neiman 2020). I explicit this relationship using the UKHLS data by plotting occupations by their median income and flexibility index. I do this exercise separately by gender to illustrate the difference in the type of

occupation both genders occupy.

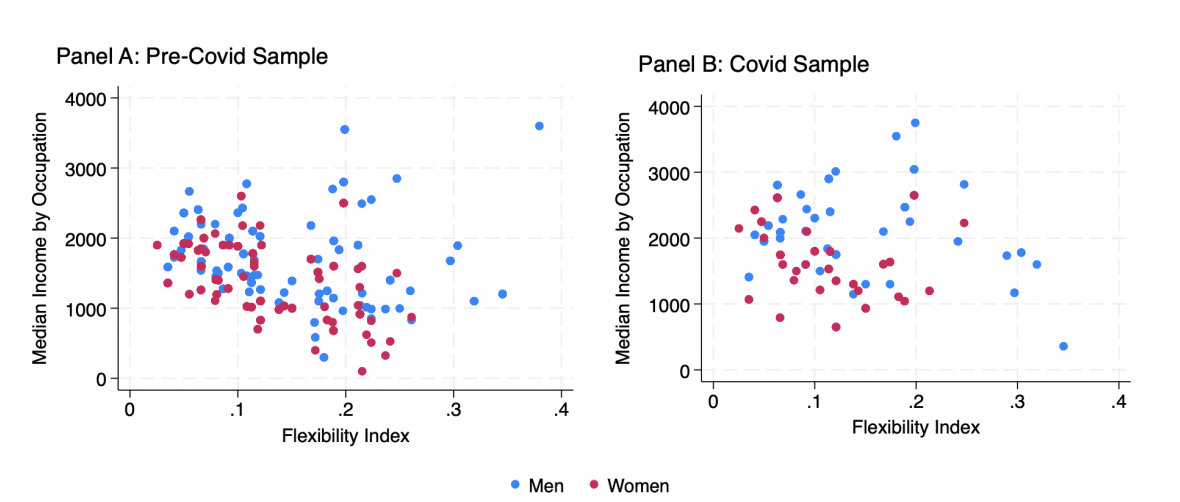


Figure 4: Median Income by Occupation over the flexibility index for Pre and Post Covid Samples

Note: Only observations 1 year before childbirth are included. Panel A includes 680 women and 559 men. Panel B includes 76 men and 105 women

In general, I find that women tend to be in jobs that are in the middle of the index, while men tend to aggregate in the extremes. Moreover, while it is true that flexible jobs are correlated with higher wages across genders, inflexible jobs are not necessarily correlated with lower wages for men. The outlier occupation on the top right corner of the pre covid sample is "Mining and mineral-processing plant operators", this occupation is the most inflexible and highest paying in the pre-COVID sample. To summarize, the flexibility index alone gives us an indication that the type of occupation the parent occupied pre-childbirth could mitigate somewhat the child penalties, this is true for men and women. However, due to a correlation with income, we are unable to fully disentangle the channel that flexibility operates through. To that end, looking at child penalties using solely the job flexibility index might give biased interpretation which motivated my approach to create new groupings combining income and flexibility.

# 6 Child penalties by flexibility and income

In this section, I focus on new groupings based on a combination of income and flexibility measure. The low-income group includes those below the 50th percentile of the income distribution, while the low-flexibility group includes those in the bottom half of the flexibility index. I then estimate child penalties for income, participation, hours worked and wage rates on the pre-COVID sample. In regards to the new groupings by income and flexibility, Table 2 shows the proportion of mothers and fathers in each group. I find unsurprisingly, that the flexible and upper income group and the inflexible and lower income group are overrepresented. Also, fathers are more present in inflexible and higher income occupations as was observed in Figure 4. Lastly it is interesting to see that women who are not in the two extremes are more likely to be in the flexible and low income group. As discussed in the introduction, the main difference with other flexibility measures is that the ability to work from home measures locational flexibility as opposed to temporal flexibility. I discuss limitations to this approach in the discussion section. Research looking at temporal flexibility focuses on part-time penalties in different occupations. Using the UKHLS measure of part time and full time employment, I find that women in the low income group are more likely to be employed part-time and those in the flexible occupation are less likely to be employed part-time. My point is that being in a locational flexible group does not necessarily correlate with temporal flexibility. More information on that distribution can be found in the appendix.

| Quartile | Mothers | Fathers | Total |
|---|---|---|---|
| 1 (inflexible and low income) | 34.90 | 23.94 | 30.00 |
| 2 (inflexible and high income) | 14.14 | 27.09 | 19.93 |
| 3 (flexible and low income) | 18.73 | 11.97 | 15.70 |
| 4 (flexible and high income) | 32.23 | 37.01 | 34.37 |

Table 2: Proportion of mothers and fathers in each quartile of the new index which combines income and flexibility

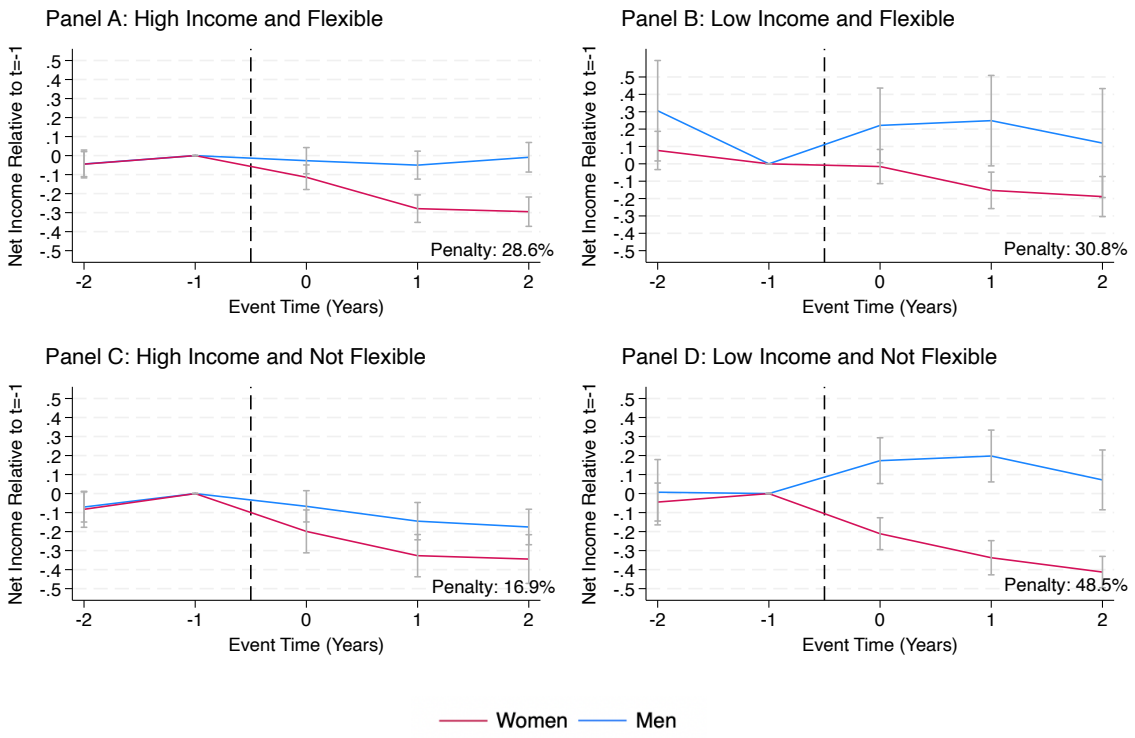


Figure 5: Child penalty in income by flexibility and income index for Pre Covid Sample

Note: Panel A observations include 1020 observations for men and 1040 for women. Panel B has 345 observations for men, 665 for women. Panel C has 710 observations for men and 474 for women. Lastly, Panel D has 720 observations for men, 1220 for women.

Figure 5 looks at child penalties in income using the new groupings. Although these new child penalties are very noisy, they provide clearer insights into the role of flexibility independent of income. The results previously observed are retained, in that less flexible jobs accrue higher child penalties. Additionally, it is now possible to differentiate between types of flexible jobs. Here women that see the smallest decline in their monthly labour income are those in the flexible and low income group. This group has a 20% penalty after 2 years compared to 30% for the high income flexible group. Interestingly, this group also shows less decline in income at birth, a trend previously observed in figure 1. Since I am not controlling for marital status, a possible explanation for this considers that women in this group might be working if they are

single mothers. Women in high income groups might be able to take maternity leave while women in lower income jobs cannot.

The fathers' side show surprising results. In particular, men in low income jobs, regardless of the flexibility measure show an increase in labour income of up to 25% one year post birth in the low income, flexible occupation group. This could point to a compensating story, that is men work more or earn more to compensate for lower income from their partner. The same mechanism could operate on the women's side. Then, women in more flexible occupations, and are able to work, would see less of a child penalty. In fact, comparing low income mothers with high flexibility to those with low flexibility reveals a 20% point difference. Since men's incomes are also responding to childbirth, child penalties show that those in the high income group converge more towards men.

An important distinction is made here between the compensating versus the comparative advantage theory. In the comparative advantage theory, the partner with the comparative advantage in the household incurs the child penalty as a result of household specialization. Many papers have disproved this theory (Kleven, Landais, and Søgaard 2021, Andresen and Nix 2022, Cortés and Pan 2023). While I cannot see these mechanisms here since I am not matching couples. We should be seeing the opposite of what we are seeing here, in that those with the high income should see less of a penalty in both genders, however I find that those in the low income flexible group show less penalties in both genders. I discuss more on matching in the discussion section. In order to find the channels through which parents' income changes we look at child penalties in hours worked, participation and wage rates.

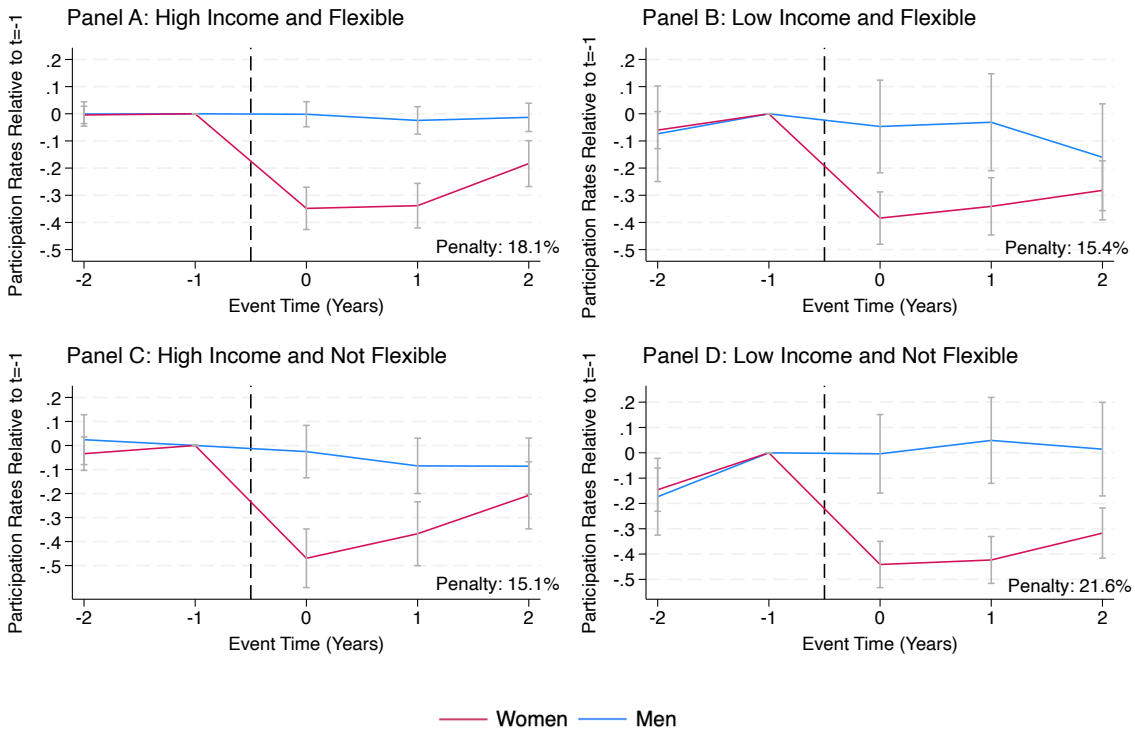


Figure 6: Child penalty in participation rates by flexibility index and income for Pre Covid Sample

Note: Panel A observations include 1020 observations for men and 1040 for women. Panel B has 345 observations for men, 665 for women. Panel C has 710 observations for men and 474 for women. Lastly, Panel D has 720 observations for men, 1220 for women.

Figure 6 shows child penalties in terms of participation rates. The event studies show relatively similar results to what we had previously seen in figure 1. Mothers see a drop in participation of around 40% at childbirth. The figure also shows that mothers in the flexible occupations see less of an initial drop in participation at childbirth, although it is less significant than the difference seen in figure 1. As in figure 1, participation rates increase in all of our groups after birth. However, women in the high income groups reenter the labour force more and quicker than those in the low income group. Men do not see significant changes in their labour force participation, which rules out the extensive margin in explaining the increase in labour income in lower income groups.

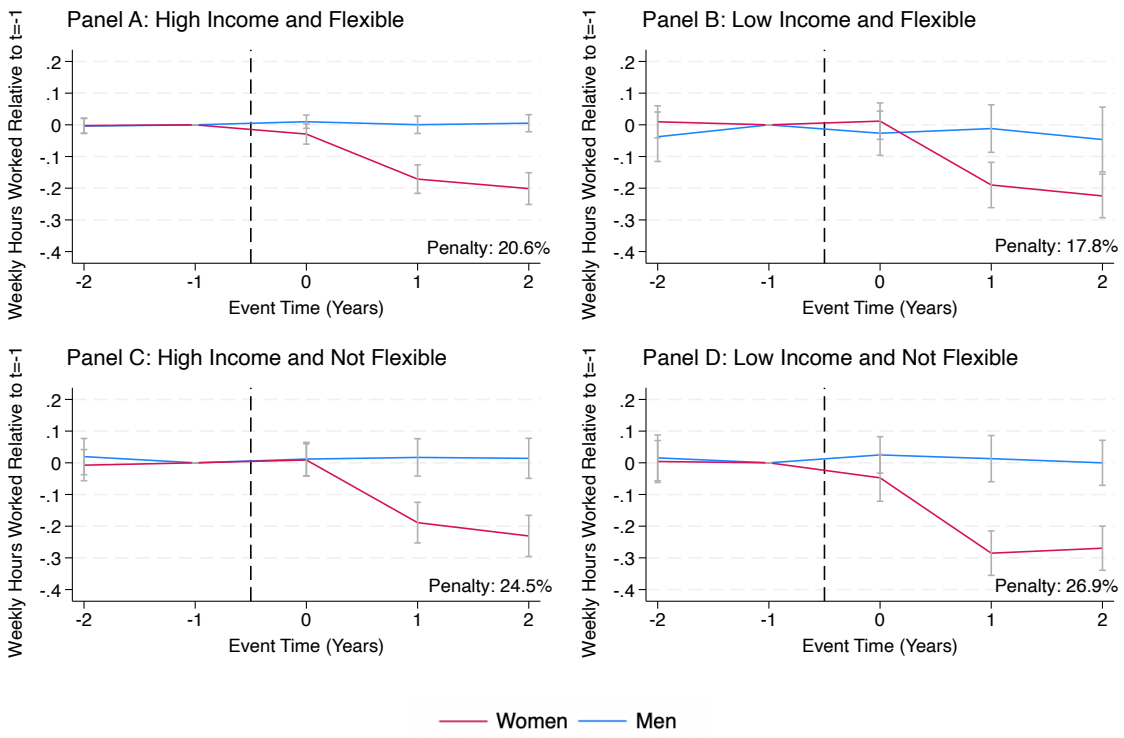


Figure 7: Child penalty in hours worked by flexibility index and income for Pre Covid Sample

Note: Panel A observations include 962 observations for men and 826 for women. Panel B has 276 observations for men, 500 for women. Panel C has 595 observations for men and 355 for women. Lastly, Panel D has 502 observations for men, 777 for women.

The second channel through which child penalties operate is through changes in hours worked (figure 7). Here, I find more significant estimates since there is less noise from those out of the labour force. Women in all four groups experience similar declines in hours worked after childbirth. However, those in the low-income, low-flexibility group have a 10% point greater decrease compared to the other groups. Thus, as seen repeatedly, flexibility seems to have a stronger impact on mothers in the lower income group. Since men show no significant changes in their hours worked, the relatively lower child penalty in income observed by the low income high flexibility group (figure 5) must come from the last channel: wages.

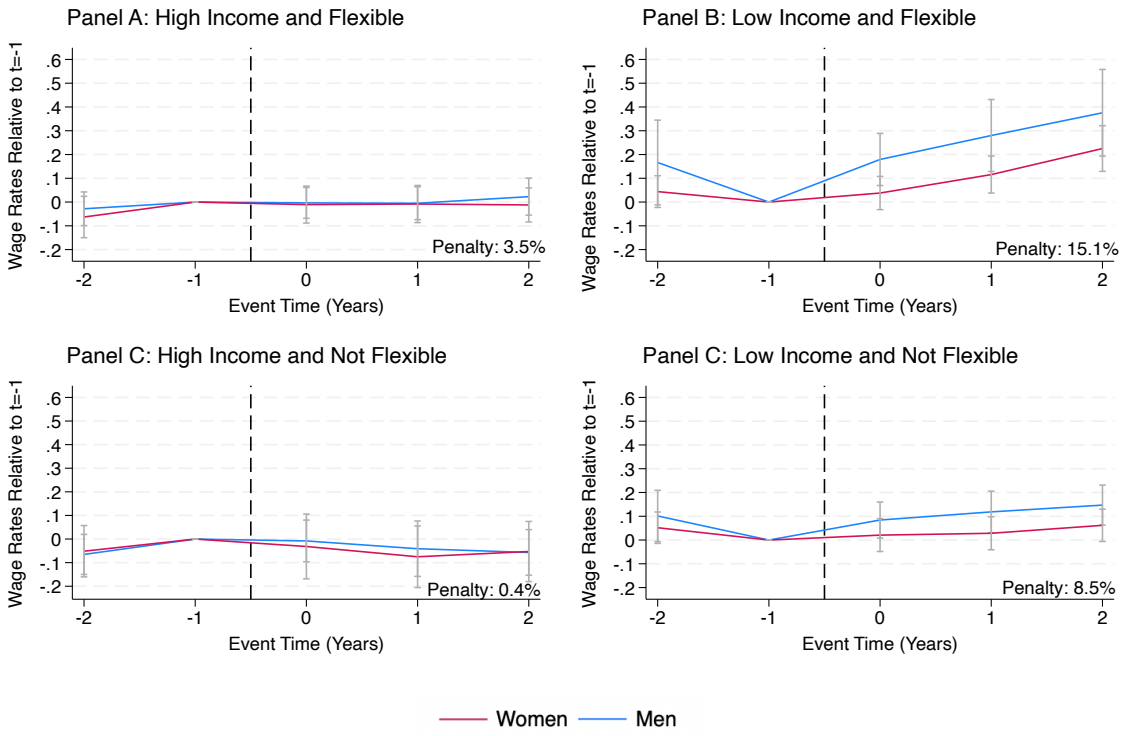


Figure 8: Child penalty in wage rates by income and flexibility for Pre Covid Sample

Note: Panel A observations include 957 observations for men and 815 for women. Panel B has 273 observations for men, 497 for women. Panel C has 594 observations for men and 355 for women. Lastly, Panel D has 497 observations for men, 737 for women. The difference with the hours worked set of observations comes from limiting the sample to above 10hours worked.

Finally, figure 8 shows the event studies in terms of wage rates. As expected, I find that men in the low income occupations see an increase in their wage rates following birth. This coincides with the increase in income we had seen in figure 5. Women in the low income flexible group also show an increase in their wages rates, suggesting they switch to higher paid jobs following birth. Research has looked at the switching of occupations in mothers following birth and finds that women tend to move to family friendly occupations Kleven, Landais, and Søgaard 2019a. While I cannot see that in my event studies, I also do not find that the high income women see a decline in their wage rates. So while flexibility does seem to have a mitigating effect on child penalties in women, the effect for the low income group seems to be primarily driven by changes in wage rates which suggests not that women are more able to work, but rather that they are switching to higher paid jobs following childbirth. This is also what we find in

men in the low income groups.

An alternative, more pertinent explanation for why low income women see an increase in wage rates after birth looks at the pre-birth period. In this explanation, low income pregnant women could be passed up for promotion while high income women are not, maybe due to better employment protections since low income women are more likely to be employed part time. In this story, the rise in wages post-childbirth is more of a 'return to baseline' rather than an increase in wages. This is supported by the fact that in time t=-2 (pre-pregnancy), wages seem to be higher for the low income women than in time t=-1 (during pregnancy).

There are two possible confounders to these explanations. One possibility is that the breakdown by flexibility and income created groups with different age distributions. Since income is correlated with age, younger people might be overrepresented in the low-income groups. Then, despite controlling for age, there might be contamination through heterogenous treatment effects that are a reflection of the age distribution (Sun and Abraham 2021, Goodman-Bacon 2021). To test this, I examined the age distribution within my categories. Although I find that the groups are not balanced in terms of age, the high-income, low-flexibility and low-income, high-flexibility groups are similar in age distribution but differ in wage increases. If heterogeneous effects were due to solely to different age distributions, I argue that the wage effects should appear in both groups. More detail on age distributions can be found in the appendix. If this is indeed the case, I could have used a stacked DiD approach with rolling control groups like the one presented in the paper by Melentyeva and Riedel although this approach might not be suitable due to my small sample (Melentyeva and Riedel 2023). Another concern is that the there might be a policy which affects the low income group but not the high income group. While the within group comparison is still possible, the comparison across groups is then limited. In other words, the time dummies should control for time trends within the concerned group but not across. One possible policy is are UK minimum wage policies that have continuously increased the real minimum

wage for 25 years (Machin 2024). Since a high proportion of individuals in the lower income category are earning the minimum wage, I cannot completely rule out this possibility. Although the difference between the low income flexible and non flexible groups suggests there might be another factor at play.

# 7 Discussion

In this section, I explore potential expansions of the analysis, including matching couples based on their partner's income and extending the analysis to the COVID period.

First, the original child penalty approach does not match couples since the effects are aggregated for all couples. By disaggregating individuals when applying the flexibility index, the analysis changes. I now match fathers and mothers in the same occupation/income group. This method isn't fully representative of real couples, as partners might differ in occupation or income category. I test this by matching couples using their personal identifiers and their partner's but lose many observations in the process since my original dataset included those who were divorced, unmarried or had missing partner ids. After matching couples in the pre-COVID sample using their observation 1 year before birth I am left with 375 couples. The correlation coefficients for these couples were 0.3537 for labor income, 0.2833 for the flexibility index, and 0.1200 for the combined flexibility and income index. Thus, while couples are somewhat similar in the metrics analyzed, they do not necessarily appear in the same group used for the analysis. As mentioned in the paper, the interpretation then changes when comparing penalties between men and women in the same group. Research on MBA graduates has shown that women with high-earning spouses are more than twice as likely not to work following childbirth compared to those with lower-earning spouses (Bertrand et al. 2010). A similar analysis could have been done to look at the likelihood of working based on the partner's flexibility index.

Additionally, it would be interesting to understand how couples switch between occupations after childbirth. Since my analysis only uses the occupation 1 year before birth, I lose out on this aspect. Investigating whether flexibility predicts less occupational switching would have been interesting, as women often move to more child friendly occupations after having a child (Kleven, Landais, and Søgaard 2019a). While my study found that flexibility predicted lower hours penalties, the findings in wage rates suggest switching is very likely. Alternatively, I would examine whether job flexibility influences how couples divide childcare and domestic labor. A serious limitation to my analysis comes from the fact that I look at locational flexiblity and not temporal flexibility. While it is justified as COVID primarily precipitated this change, it is less pertinent since childcare is still required during working hours. In other words, men and women working at home are still working, and childcare is a demanding task that cannot often be done simultaneously. While locational flexibility offers some time savings, it cannot fully meet the demands of childcare. Future research could look at both flexibility measures looking at their respective impacts on child penalties.

Lastly, given more years and observations I would have extended this analysis to the COVID sample. Since I did not have enough data points, separating my sample by occupation and income, gave inconclusive estimates. Ideally, I would have compared child penalties across the two samples. The story being that those in the flexible occupations in the COVID sample would see less penalties than those in the pre-COVID sample due to changes in work from home arrangements. Initial findings hint at this direction (see Figure 1 and Figure 2), but large standard errors prevent conclusive comments.

# 8 Conclusion

In this paper, I set out to look at the effect of occupation flexibility on child penalties, in hopes to find if the pandemic, which brought about new work arrangements, changed the persistent nature of these penalties. I first showed that child penalties in terms of income, hours worked, participation, and wage rates have changed before and after COVID. Despite the noisy estimates, this motivated me to consider a flexibility measure based on the ability to work from home.

I introduced this measure into my dataset and computed child penalties on the pre-COVID sample only, as the COVID sample was too small for reliable event studies. I found that women's income dropped less in flexible occupations, while men's income in less flexible occupations also dropped. This unexpected result prompted me to explore the relationship between income and flexibility, leading me to create a new index that combined both factors to better understand their effects separately.

The new penalties indicated that men in low-income groups saw an increase in labor income after birth, while women consistently experienced an income drop. Amongst women, three groups emerged: women with high income experienced similar penalties regardless of flexibility, women in the low-income inflexible occupations showed the greatest income drop, and women in the low-income flexible occupation group which had the smallest decline.

Further analysis revealed that differences in penalties were primarily due to wage rates rather than hours worked or participation which suggested occupation switching rather than a flexibility related mechanism. Although, I did find that mothers in low income flexible occupations saw less of a decline in hours worked compared to women in the low income low flexibility group, suggesting that flexibility might have a protective effect on women with lower income. I found that women and men in low-income, flexible occupations saw wage rate increases following birth and proposed three ways to explain this increase. I finally discussed how the analysis could be expanded for future research.

Research on child penalties offer insights on how parents' careers diverge after childbirth. Since the divergence is explained by the parents' choice, this does not necessarily call of government intervention. However, by disaggregating impacts and following individuals in similar occupations, I find that women, especially those in low-income, inflexible jobs, consistently fare worse in the labor market. Although most child penalty research does not offer policy suggestions, my findings imply that affordable childcare targeted at these occupations could reduce child penalties for this population.

# 9 Appendix

| Occupation (isco88 code) | Pre-Covid | Covid | Flexibility Index |
|---|---|---|---|
| Personal care and related workers (513) | 10.23 | 5.13 | 0.15 |
| Other departmental managers (123) | 5.56 | 11.11 | 0.06 |
| Finance and sales associate professionals (341) | 5.05 | 11.97 | 0.07 |
| Other office clerks (419) | 4.29 | 2.56 | 0.04 |
| Primary and pre-primary education teach (233) | 4.17 | 4.27 | 0.10 |
| Shop salespersons and demonstrators (522) | 3.79 | 2.56 | 0.12 |
| Production and operations department manager (122) | 3.66 | 3.42 | 0.11 |
| Business professionals (241) | 3.41 | 0.85 | 0.05 |
| Nursing and midwifery associate professions (223) | 3.41 | 4.27 | 0.17 |
| Modern health associate professionals (322) | 3.16 | 3.42 | 0.17 |
| Client information clerks (422) | 2.53 | 3.42 | 0.12 |
| Secondary education teaching profession (232 ) | 2.90 | 5.13 | 0.09 |
| Domestic and related helpers, cleaner (913) | 1.39 | 3.42 | 0.19 |
| Health professionals (except nursing) (222) | 1.64 | 3.42 | 0.20 |
| Missing | 1.26 | 1.71 | |
| Inapplicable | 11.87 | 6.84 | |
| Total numer of occupations | 64 | 31 | |
| Total sample of mothers | 792 | 117 | |

(a) Top non-Covid and Covid Sample Mother's occupation (1y before birth)

| Occupation (isco88 code) | Pre-Covid | Covid | Flexibility Index |
|---|---|---|---|
| Computing professionals (213) | 5.30 | 7.87 | 0.07 |
| Other departmental managers (123) | 7.22 | 5.62 | 0.06 |
| Production and operations department managers (122) | 5.62 | 4.49 | 0.11 |
| Finance and sales associate professionals (341) | 5.30 | 7.87 | 0.07 |
| Architects, engineers and related professions (214) | 4.98 | 4.49 | 0.05 |
| Health professionals (except nursing) (222) | 1.77 | 4.49 | 0.20 |
| Other office clerks (419) | 4.29 | 3.37 | 0.04 |
| Artistic, entertainment and sports associate (347) | 4.01 | 4.49 | 0.12 |
| Business professionals (241) | 2.41 | 3.37 | 0.05 |
| Shop salespersons and demonstrators (522) | 2.09 | 3.37 | 0.12 |
| Protective services workers (516) | 1.12 | 3.37 | 0.19 |
| Missing | 0.96 | 5.62 | |
| Inapplicable | 8.03 | 8.99 | |
| Total numer of occupations | 78 | 34 | |
| Total sample of fathers | 623 | 89 | |

(b) Top non-Covid and Covid Sample Father's occupation (1y before birth)

Table 3: Top Occupations and Associated Index

| Occupation | Mothers | Fathers |
|---|---|---|
| 1 (Most Flexible) | Other departmental managers | Architects, engineers and related profe |
| | Other office clerks | Other departmental managers |
| | Business professionals | Computing professionals |
| 2 | Secondary education teaching profession | Secondary education teaching profession |
| | Primary and pre-primary education teach | Numerical clerks |
| | Finance and sales associate professiona | Finance and sales associate professiona |
| 3 | Personal care and related workers | Shop salespersons and demonstrators |
| | Shop salespersons and demonstrators | Artistic, entertainment and sports asso |
| | Production and operations department ma | Production and operations department ma |
| 4 (Least flexible) | Housekeeping and restaurant services | Health professionals (except nursing) |
| | Nursing and midwifery associate profession | Machinery mechanics and fitters |
| | Modern health associate professionals | Building finishers and related trades w |

Table 4: Top occupations by gender and flexibility index

| Men | | | | | |
|---|---|---|---|---|---|
| | Age group | | | | |
| Quartile | <22 | 23-27 | 28-32 | 33-37 | 38+ |
| 1 (inflexible +low income) | 10.53 | 21.05 | 36.84 | 19.74 | 11.84 |
| 2 (inflexible +high income) | 0.62 | 11.25 | 34.38 | 31.25 | 22.50 |
| 3 (flexible +low income) | 4.00 | 14.67 | 36.00 | 28.00 | 17.33 |
| 4 (flexible +high income) | 0 | 3.95 | 35.53 | 39.47 | 21.05 |
| Women | | | | | |
| 1 (inflexible +low income) | 20.75 | 33.21 | 29.06 | 13.58 | 3.40 |
| 2 (inflexible +high income) | 0 | 11.82 | 45.45 | 34.55 | 8.18 |
| 3 (flexible +low income) | 4.20 | 26.57 | 43.36 | 19.58 | 6.29 |
| 4 (flexible +high income) | 0 | 4.98 | 49.79 | 37.34 | 7.88 |

Table 5: Proportion of mothers and fathers in each quartile of the flexibility/income index by age group for the pre-covid sample

| Quartile | Full-time | Part-time | Missing |
|---|---|---|---|
| Women | | | |
| 1 (inflexible +low income) | 72.08 | 24.15 | 3.77 |
| 2 (inflexible +high income) | 95.45 | 3.64 | 0.91 |
| 3 (flexible +low income) | 84.62 | 13.99 | 1.40 |
| 4 (flexible +high income) | 96.27 | 2.07 | 1.66 |
| Men | | | |
| 1 (inflexible +low income) | 84.87 | 7.24 | 7.89 |
| 2 (inflexible +high income) | 95.62 | 1.88 | 2.50 |
| 3 (flexible +low income) | 85.33 | 8.00 | 6.67 |
| 4 (flexible +high income) | 94.74 | 0.44 | 4.82 |

Table 6: Proportion of mothers and fathers in part time or full time employment by quartile for the precovid sample

# Works Cited


**Aksoy**, **Cevat Giray**, et al. "Time Savings When Working from Home". *AEA Papers and Proceedings*, vol. 113, May 2023, pp. 597–603. https://doi.org/10.1257/pandp.20231013.

**Andresen**, **Martin Eckhoff**, and **Emily Nix**. "What causes the child penalty? Evidence from adopting and same-sex couples". *Journal of labor economics*, vol. 40, no. 4, 2022, pp. 971–1004.

**Andrew**, **Alison**, et al. "Women and men at work". *IFS Deaton Review of Inequalities*, 2021.

**Benny**, **Liza**, et al. *Occupation flexibility and the graduate gender wage gap in the UK*. ISER Working Paper Series, 2021.

**Bertrand**, **Marianne**, et al. "Dynamics of the gender gap for young professionals in the financial and corporate sectors". *American economic journal: applied economics*, vol. 2, no. 3, 2010, pp. 228–55.

**Collins**, **Caitlyn**, et al. "COVID-19 and the gender gap in work hours". *Gender, Work & Organization*, vol. 28, 2021, pp. 101–12.

**Cortés**, **Patricia**, and **Jessica Pan**. "Children and the Remaining Gender Gaps in the Labor Market". *Journal of Economic Literature*, vol. 61, no. 4, Dec. 2023, pp. 1359–409. https://doi.org/10.1257/jel.20221549.

**Davis**, **Steven J**. *The Big Shift in Working Arrangements: Eight Ways Unusual*. National Bureau of Economic Research, 2024.

**De Quinto**, **Alicia**, et al. "The child penalty in Spain". *Banco de Espana Occasional Paper*, no. 2017, 2020.

**Dingel**, **Jonathan I**, and **Brent Neiman**. "How many jobs can be done at home?" *Journal of public economics*, vol. 189, 2020, p. 104235.

**Fairlie**, **Robert W**, et al. *The evolving impacts of the covid-19 pandemic on gender inequality in the us labor market: The covid motherhood penalty*. National Bureau of Economic Research, 2021.

**Ferrant**, **Gaëlle**, et al. "Unpaid Care Work: The missing link in the analysis of gender gaps in labour outcomes". *Boulogne Billancourt: OECD Development Center*, vol. 20, 2014.

**Goldin**, **Claudia**. "A Grand Gender Convergence: Its Last Chapter". *American Economic Review*, vol. 104, no. 4, Apr. 2014, pp. 1091–119. https://doi.org/10.1257/aer.104.4.1091.

**Goldin**, **Claudia**, and **Lawrence F Katz**. "A most egalitarian profession: pharmacy and the evolution of a family-friendly occupation". *Journal of Labor Economics*, vol. 34, no. 3, 2016, pp. 705–46.

**Goodman-Bacon**, **Andrew**. "Difference-in-differences with variation in treatment timing". *Journal of econometrics*, vol. 225, no. 2, 2021, pp. 254–77.

**Gottlieb**, **Charles**, et al. "Working from home in developing countries". *European Economic Review*, vol. 133, 2021, p. 103679.

**Hardy**, **W**. "Occupation classifications crosswalks–from O* NET-SOC to ISCO". *Institute for Structural Research (IBS), Warsaw, Poland. Available online: https://ibs.org. pl/en/resources/occupationclassifications-crosswalks-from-onet-soc-to-isco*, 2016.

**Kleven**, **Henrik**, **Camille Landais**, and **Gabriel Leite-Mariante**. *The child penalty atlas*. National Bureau of Economic Research, 2023.

**Kleven**, **Henrik**, **Camille Landais**, **Johanna Posch**, et al. "Child Penalties across Countries: Evidence and Explanations". *AEA Papers and Proceedings*, vol. 109, May 2019, pp. 122–26. https://doi.org/10.1257/pandp.20191078.

**Kleven**, **Henrik**, **Camille Landais**, and **Jakob Egholt Søgaard**. "Children and Gender Inequality: Evidence from Denmark". *American Economic Journal: Applied Economics*, vol. 11, no. 4, Oct. 2019, pp. 181–209. https://doi.org/10.1257/app.20180010.

———. Replication data for: Children and gender inequality: Evidence from Denmark. 2019.

———. "Does biology drive child penalties? evidence from biological and adoptive families". *American Economic Review: Insights*, vol. 3, no. 2, 2021, pp. 183–98.

**Liu**, **Jiaqi**. "Racial difference in child penalty". 2021.

**Machin**, **Stephen**. "Wage controversies: real wage stagnation, inequality and labour market institutions". *LSE Public Policy Review*, vol. 3, no. 2, 2024.

**Melentyeva**, **Valentina**, and **Lukas Riedel**. *Child penalty estimation and mothers' age at first birth*. ECONtribute Discussion Paper, 2023.

**O*NET**. O*NET 20.1 Data" O*NET Resource Center, National Center for O*NET Development. 2015. www.onetcenter.org/dictionary/20.1/excel/.

**Sébastien Fontenay**, **Thomas Murphy**, and **Ilan Tojerow**. "Child penalties across industries: why job characteristics matter". *Applied Economics Letters*, vol. 30, no. 4, 2023, pp. 488–95. *https://doi.org/10.1080/13504851.2021.1994518*, https://doi.org/10.1080/13504851.2021.1994518.

**Sun**, **Liyang**, and **Sarah Abraham**. "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects". *Journal of econometrics*, vol. 225, no. 2, 2021, pp. 175–99.

**University of Essex, Institute for Social and Economic Research**. Understanding Society: Waves 1-13, 2009-2022 and harmonised BHPS: Waves 1-18, 1991-2009. 2023. https://doi.org/http://doi.org/10.5255/UKDA-SN-6614-19.